Bill of Materials

RescueSwarm: An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Three-aircraft programme · Revision 3 · 21 September 2026

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Accompanies The programme brief and its phase-wise approval letters

1. Purpose

This document lists every component needed to build, test and operate the three RescueSwarm aircraft and their ground station, with quantities, unit prices and the supplier each price was taken from. Every price is from a dated Indian listing, and each supplier name in Section 4 links to it. Two lines have no listing and are marked as estimates. The totals are the same as those in the programme brief and the approval letters, because all three are generated from one parts list.

2. Cost summary

Parts at retail, GST already included (56 lines)	4,79,017
Parts quoted before GST, and estimates (5 lines)	14,961
Parts total, three aircraft (61 lines)	**4,93,978**
GST at 18 % on the pre-GST lines only	2,693
Contingency, 15 %	74,501
Total requested	**INR 5,71,171**

The components fitted to one aircraft cost INR 1,37,509. The request is released in 29 phases, each only after the previous one has produced its result. The first release is INR 3,733, and nothing above INR 26,000 is spent until a motor has been measured on the thrust stand.

Group	Lines	Per aircraft	All three
Test instruments	6	–	3,246
Avionics, sensing and RTK positioning	22	72,317	2,44,677
Airframe, propulsion and power	19	65,192	2,03,243
Flight-test support	7	–	29,867
Ground station	7	–	12,945
Total	**61**	**1,37,509**	**4,93,978**

3. System configuration

Airframe	Quadrotor, 18 in carbon propellers, Tarot 5008 340 KV motors, carbon-tube arms; about 7 kg take-off mass with four relief kits
Flight control	Holybro Kakute H7 V1.5 running ArduPilot, stacked on a 4-in-1 65 A speed controller
Onboard computing	Raspberry Pi 5 (4 GB) with AI HAT+ (26 TOPS) for person detection; 12.3 MP camera, 6 mm lens
Positioning	Centimetre-level RTK on every aircraft (ZED-F9P, dual-band, with compass); own ground base station, so no internet is needed
Video	Digital downlink (wfb-ng) carrying the camera feed with detections drawn on; one directional ground receiver per aircraft
Links	2.4 GHz ExpressLRS for the safety pilot; 868 MHz LoRa for RTK corrections
Power	6S 16 Ah LiHV lithium-polymer pack; separate regulators for the computer, the video radio and the servos

4. Bill of materials

All prices are in INR. *Per AC* is the number fitted to one aircraft; a dash means the item is bought once for the whole programme. *Qty* is the programme quantity, including spares.

#	Item	Part	Per AC	Qty	Unit	Amount	Source
Test instruments							**3,246**
1	Load-cell amplifier	7Semi HX711 24-bit ADC	–	1	104	104	Robu
2	Load cell	REES52 20 kg load cell + HX711 breakout	–	1	854	854	Amazon India
3	Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	–	1	373	373	Flipkart
4	Fasteners	Hex socket-head assortment M3/M4/M5, SS304	–	1	1,140	1,140	OnlyScrews
5	Threadlocker	Loctite 243, 50 ml	–	1	664	664	Amazon India
6	Throttle source	Servo tester / ESC consistency tester, PPM	–	1	111	111	Robokits
Avionics, sensing and RTK positioning							**2,44,677**
7	Flight controller	Holybro Kakute H7 V1.5, no Bluetooth	1	3	9,269	27,807	Indian Robo Store
8	FC vibration mount	Glass-fibre anti-vibration shock absorber	1	3	161	483	Robu
9	Autopilot log card	SanDisk 32 GB microSDHC, class 10	1	3	1,549	4,647	Robocraze
10	Companion computer	Raspberry Pi 5, 4 GB	1	3	11,049	33,147	Yantronix
11	AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	3	11,749	35,247	Robu
12	Compute cooling	Official Raspberry Pi 5 active cooler	1	3	519	1,557	Robu
13	Storage	SanDisk High Endurance 128 GB microSD	1	3	3,989	11,967	Flipkart
14	GNSS RTK rover	GenX F9P Air: ZED-F9P L1/L2 + helix + QMC5883 compass, ∼20 g	1	3	15,392	46,176	Robokits
15	GNSS RTK base	SmartElex ZED-F9P-04B RTK board, U.FL	–	1	17,965	17,965	Robocraze
16	GNSS antenna, base	LS-125-A L1/L2/L5 survey antenna	–	1	8,123	8,123	Robokits
17	GNSS feeder, base	U.FL to SMA female pigtail, 10 cm	–	1	49	49	Robocraze
18	GNSS cable, base	RG58 50 ohm, 3 m, TNC-M to SMA-M, made locally	–	1	600	600	Estimate[†]
19	RTK base radio	EByte E22-900T22D-V2, SX1262, 22 dBm	–	1	989	989	Robu
20	Height rangefinder	Benewake TFmini-S, 12 m lidar, UART	1	3	2,827	8,481	Indian Robo Store
21	Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	3	6,799	20,397	Robu
22	Video transmitter	LB-Link BL-M8812EU2, RTL8812EU, 5 GHz 2T2R	1	3	1,849	5,547	Hubtronics
23	Video antennas	6 dBi 5 GHz dipole + 165 mm U.FL lead	2	6	75	450	Robocraze
24	Command receiver	ExpressLRS RX24T, 2.4 GHz	1	3	1,439	4,317	Robu
25	Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	3	989	2,967	Robu
26	BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3	3,939	11,817	Amazon India

continued on next page

#	Item	Part	Per AC	Qty	Unit	Amount	Source
27	BEC, video radio	Hobbywing 3 A UBEC, 5/6 V, 5 A peak, 2-6S	1	3	369	1,107	KitsGuru
28	BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	3	279	837	Robu
Airframe, propulsion and power							**2,03,243**
29	Motors	Tarot TL96020, 5008, 340 KV	4	12	4,703	56,436	Robokits
30	Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	4	16	1,641	26,256	Robokits
31	Speed controllers	Hobbywing XRotor FPV G2, 4-in-1, 65 A, 3-6S	1	3	6,500	19,500	RC Mumbai
32	Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	6	2,259	13,554	Robu
33	Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm	1	3	2,895	8,685	Indian Robo Store
34	Motor mounts	Tarot 25 mm motor mount, TL9602	4	12	1,119	13,428	Robu
35	Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3	3,012	9,036	Amazon India
36	Landing gear	F450/F550 landing skid	1	3	549	1,647	Robu
37	Suspension springs	5 mm stainless double torsion spring	10	32	2	64	IndiaMART
38	Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	–	1	949	949	Robu
39	Release servos	MG90S mini servo, 180 deg	4	12	149	1,788	Robocraze
40	Battery pack	GenX 22.8V 6S 16000mAh 25C/40C HV LiPo, AS150	1	3	12,197	36,591	Robokits
41	Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3	3,000	9,000	Njour
42	Pack retention	300 mm LiPo strap, reusable	2	6	62	372	Robu
43	Main leads	10 AWG ultra-flexible silicone wire	–	1	149	149	Robu
44	Power connectors	Amass XT90S anti-spark, M/F pair	4	12	180	2,160	Robokits
45	Signal connectors	JST ZH 6-pin, 200 mm leads	1	3	23	69	Robu
46	Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	3	1,139	3,417	Desertcart
47	Insulation	PVC heat-shrink sleeve, 150 mm	1	3	47	141	Robu
Flight-test support							**29,867**
48	Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	–	1	16,519	16,519	zBotic
49	Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	–	1	7,349	7,349	Robu
50	Cable management	Nylon zip ties, 350 mm, 100 pcs	–	1	135	135	Robu
51	Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	–	1	159	159	Robu
52	Mounting tape	3M VHB 5952, 6 mm x 8.2 m	–	1	560	560	Amazon India
53	Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	–	1	145	145	Amazon India
54	Consumables	Fasteners, tape, connectors, filament	–	1	5,000	5,000	Estimate†
Ground station							**12,945**
55	Video receivers	LB-Link BL-M8812EU2, RTL8812EU, one per aircraft channel	–	3	1,849	5,547	Hubtronics
56	Receive panels	5.8 GHz 14 dBi directional panel, RP-SMA, 62x55 mm	–	3	566	1,698	FlyRobo
57	Panel pigtails	U.FL to RP-SMA female pigtail, RG178	–	3	99	297	IndiaMART
58	Receive antennas	6 dBi 5 GHz dipole + U.FL lead, second port	–	3	75	225	Robocraze
59	Receiver USB hub	Transcend HUB3 powered 4-port USB hub + adapter	–	1	2,653	2,653	Flipkart
60	Coordination base	EByte E22-900T22U, SX1262, USB	–	1	1,646	1,646	Hubtronics
61	Base station mount	Photographic tripod and adapter	–	1	879	879	Amazon India
Parts total, three aircraft							**4,93,978**

† No listing. The base antenna cable is to be made up by a local RF cable shop, and consumables are a round allowance. Both are carried as pre-GST and have GST and contingency applied.

5. Battery selection

The battery is the largest single design decision in this revision. The previous design used a pack built in-house from eighteen cells. A bought lithium-polymer pack avoids spot-welding cells in a student lab. It has to be large enough to keep the mission's energy reserve *after* its own extra weight is counted, because a heavier aircraft needs more power to hover. Each candidate was run through the project's sizing model at its own mass:

Pack	Mass kg	Energy Wh	Take-off mass, kg	Thrust/ weight	Endurance ratio	Landing reserve, Wh
Li-ion 6S3P, in-house (previous)	1.45	292	6.36	2.00	2.02	+1.1
LiPo 12 Ah	1.42	266	6.34	2.01	1.85*	−3.7*
LiPo 13 Ah	1.52	289	6.43	1.98	1.96*	−0.5*
LiPo 14 Ah	1.64	311	6.55	1.94	2.06	+2.5
LiPo 16 Ah LiHV (selected)	**1.88–2.21**	**365**	**6.8–7.1**	**1.8–1.9**	**2.1–2.3**	**+5 to +9**

Endurance ratio is hover time divided by mission time; the design requires at least 2.0. *Landing reserve* is the energy left after the third aircraft waits its turn and descends, and must be positive. * Fails the requirement. The 16 Ah pack's mass is a range because the maker does not publish it.

The 12 and 13 Ah packs fail both requirements, and about 14 Ah is the smallest that passes. The 16 Ah pack is the smallest suitable size on sale in India. Its cost is weight: on the same motors, thrust-to-weight falls from 2.0 to about 1.8–1.9 unless the thrust stand measures at least 3.55 kgf per motor. That measurement is the first gate in Section 6.

6. Verification before purchase and flight

The following are checked before the relevant purchase or before first flight. Each is either a measurement this programme exists to make, or a confirmation that a listing cannot give.

#	Item	Check	Acceptance	When	Status
1	Battery mass	Weigh each pack on receipt.	Expected 1.88–2.21 kg. The measured mass replaces the estimate in the take-off mass.	On receipt	**Open**
2	Motor thrust	Thrust-stand test of one motor, propeller and speed controller.	≥ 3.55 kgf per motor, which keeps thrust-to-weight at 2.0 at the heaviest expected mass.	Phase 2	**Open**
3	Battery charging	Set the charger profile.	LiHV, 4.35 V per cell maximum.	Before first charge	**Open**
4	Battery connector	Fit mating connectors.	AS150 mates on the aircraft harness, not XT90S.	Before ordering	**Open**
5	Pack fuse	Log peak current on the stand and in the first flights.	The 150 A fuse must hold against a $\sim$136 A peak. This is the tightest electrical margin.	Phase 2	**Open**
6	Failsafe parameters	Weigh the pack, measure its voltage sag under load, then update the pack model and the failsafe parameters and re-test.	The current parameters describe the previous Li-ion pack. **No flight before this is complete.**	Before first flight	**Open**
7	Speed controller (65 A)	Thrust-stand test with 18 in propellers and 340 KV motors.	Current, temperature and freedom from desynchronisation confirmed before the remaining two are bought.	Phase 2	Candidate
8	Base RTK antenna cable	Obtain a quote from a local RF cable shop.	3 m RG58, 50 Ω, TNC-male to SMA-male. The INR 600 in this BOM is an estimate until quoted.	Before ordering	**Open**
9	Ground video antennas	Install one 14 dBi panel per receiver, aimed at its aircraft, and an omni on the second port.	Confirm polarisation before installation. Measure the range on our own link, not the listing's analog-FPV figure. Set coax lengths from the station layout.	Ground-station build	**Open**
10	USB hub	Check the supplied power adapter.	Rated for three receivers (up to 1.8 A each when transmitting).	On receipt	**Open**

#	Item	Check	Acceptance	When	Status
11	RTK rover (GenX F9P Air)	Check the maker's specification.	Dual-band: GPS L1/L2, Galileo E1/E5b, BDS B1/B2, GLONASS G1/G2; QMC5883 compass.	Before ordering	Verified

7. Basis of prices

- Prices were taken from Indian supplier listings on 19–20 September 2026. Where a supplier blocked automated checking, the same part's price at another Indian retailer was used, and that retailer is the one linked.
- Retail listings already include GST and are used as listed. Pre-GST quotations, B2B listings and estimates have 18 % GST added (INR 2,693).
- A 15 % contingency is applied to the whole, for price movement between now and each phase's purchase.
- Customs duty does not apply: every part is bought from a supplier in India.

8. Team and approval

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
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